

Shyam Ravishankar

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Education

Rose-Hulman Institute of Technology

B.S. in Mathematics, Cum Laude

Thesis on Truncated Euler Systems

Minor in Statistics

Terre Haute, IN

Completed May 2024

Publications/Preprints

- [1] William R. Green, Connor Lane, Benjamin Lyons, Shyam Ravishankar, and Aden Shaw. The massless Dirac equation in three dimensions: Dispersive estimates and zero energy obstructions. *To appear in Journal of Differential Equations.*

Presentations

1. *Special Units and Euler Systems* Terre Haute, IN
Rose-Hulman senior thesis poster presentation May 1, 2024
2. *Analyzing Partial Differential Equations with Oscillatory Integrals* (based on [1]) San Francisco, CA
Joint Mathematics Meetings AMS-PME Undergraduate Student Poster Session January 5, 2024

Conferences/Workshops

1. IAS/Park City Mathematics Institute Park City, UT
Undergraduate Summer School on Motivic Homotopy Theory July 2024
2. Connecticut Summer School in Number Theory University of Connecticut, Storrs, CT
Summer School and Conference June 2024

Relevant Coursework

Analysis/DEs: Intro to Real Analysis, Differential Forms, Measure Theory, Functional Analysis, Complex Variables, Intro to PDEs, Calculus of Variations

Algebra: Abstract Algebra, Linear Algebra I and II

Number Theory/Algebraic Geometry: Number Theory, p -adic Analysis, Elliptic Curves and Applications, Arithmetic Geometry, Scheme Theory I and II, Cohomology of Number Fields

Topology: Algebraic Topology

Mathematical Physics: Theoretical Physics Laboratory, Special Relativity, General Relativity

Probability/Statistics: Mathematical Statistics, Bayesian Data Analysis

Computer Science/Data Science: Design and Analysis of Algorithms, Deep Learning, Natural Language Processing

Work Experience

Rose-Hulman Mathematics Department	Terre Haute, IN
<i>Grader – MA376 Abstract Algebra</i>	Spring 2024
<i>Grader – MA473 Design and Analysis of Algorithms</i>	Winter 2024
<i>Course Assistant – MA470 Galois Theory</i>	Winter 2024
• Suggested homework problems and graded homework	

<i>Grader – MA366 Real Analysis</i>	Fall 2023
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Humana Insurance Company	Louisville, KY
<i>IDEA Intern</i>	May – Aug. 2022

- Deployed a sandbox storage account with private endpoints, keyvault, data factory with log analytics in Azure
- Refactored pipelines for transitioning existing data platform to multiple Azure subscriptions
- Transitioned data platform offerings to APIs for easy and controlled deployment

<i>EDW Cloud Intern</i>	June – Aug. 2020
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- Developed a Proof of Concept in Microsoft's PowerApps to load and edit data in Azure SQL Server
- Created a dashboard in Microsoft Power BI to track teams' cloud migration progress

<i>EDW Intern</i>	Jun – Aug. 2019
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- Identified Protected Health Information (PHI) fields within over 1,000 data tables on Oracle SQL Database to have monitoring activated on those tables

Volunteering

North South Foundation	Louisville, KY
<i>Pronouncer</i>	2019 – 2022
• Pronouncer for Geography Bee contests held to raise money for underprivileged children in India	

Professional Affiliations

Mathematical Association of America	2020 –
Association for Women in Mathematics	2022 –

Honors

Mathematics Department Award for best senior thesis presentation	2024
Dean's List	10 Quarters
Indiana College Mathematics Competition Winner	2024
Theodore Paine Palmer Freshman Mathematics Award	2021

Other Activities

Rose-Hulman Badminton Club	
<i>Treasurer</i>	2021 – 2024
Rose-Hulman Concert Band	
<i>Clarinetist</i>	2020 – 2024
<i>Librarian</i>	2021 – 2024

Rose-Hulman Symphony Orchestra*Clarinetist*

2020 – 2024

Soloist – Mozart Clarinet Concerto

May 2024

Rose-Hulman Drama Club*Clarinetist – Pit Orchestra*

2022 – 2024

Louisville Winds*Clarinetist*

2024 –